

SOME ASYMPTOTIC PROPERTIES OF CONSTRAINED GENERALIZED LEAST SQUARES ESTIMATION IN COVARIANCE STRUCTURE MODELS

Sik-Yum Lee

Department of Mathematics
University Science Centre
The Chinese University of Hong Kong
Hong Kong

P.M. Bentler

Department of Psychology
University of California
Los Angeles
California 90024

Key words: Asymptotic distributions, constrained estimation, covariance structures, generalized least squares, reparameterization.

SUMMARY

Some basic results by Browne (1974) on generalized least squares estimation in the analysis of covariance structures are extended to covariance structures with parameters subject to arbitrary nonlinear constraints. It is shown that the constrained estimators are consistent, asymptotically multivariate normally distributed, and asymptotically equivalent to constrained maximum likelihood estimators. Asymptotic chi-square tests are developed to evaluate appropriate model comparisons. The relationships between the Lagrangian approach and the reparameterization approach are discussed.

1. Introduction

Covariance structure analysis has become an important tool for the analysis of structured linear models of multivariate random data (cf., Anderson, (1973); Bentler (1980); Bentler & Weeks, (1980); Jöreskog, (1973), (1978); McDonald, (1978)). In such models, $\Sigma = \Sigma(\theta)$ is a $p \times p$ covariance matrix whose elements are assumed to be differentiable real-valued functions of a true though unknown $q \times 1$ vector of parameters θ , and the primary statistical problems involve estimation of parameters of the model, establishing properties of the estimators, and evaluating goodness of fit of competing models. The dominant method of estimation in such models has been the method of maximum likelihood (e.g., Jöreskog & Sörbom, (1978)). The foundation for estimation by generalized least squares was developed by Browne (1974) and integrated into iterative estimation methods that can provide either maximum likelihood or generalized least squares estimates (Lee & Jennrich, (1979)). In these developments, the parameter vector θ is considered to represent a vector of free elements that are independent of each other. Simple equality constraints among parameters can be implemented by a reparameterization (e.g., Jöreskog, (1973)), but models containing parameters that are related by general functional constraints have not been studied. Aitchison and Silvey (1959) provided the sta-

tistical basis for obtaining and evaluating constrained maximum likelihood estimates, but their methods have not been applied to covariance structure analysis. This paper provides a statistical basis for constrained generalized least squares estimators.

2. Asymptotic Results

Let $\Sigma_0 = \Sigma(\underline{\theta}_0)$ be a $p \times p$ population covariance matrix whose elements $\sigma_{ij}(\underline{\theta}_0)$, $i, j = 1, \dots, p$ are differentiable real-valued functions of a true though unknown $q \times 1$ vector of parameters $\underline{\theta}_0$. Consider a closed and bounded parameter set $\Omega \in \mathbb{R}^q$ with elements $\underline{\theta}$. Suppose that ω is a subset of Ω whose elements satisfy the functional relationship $\underline{h}(\underline{\theta}) = \underline{0}$, where $\underline{h}(\underline{\theta})$ is an $r \times 1$ real vector-valued continuous function of $\underline{\theta}$. We consider the situation where $\underline{\theta}_0$ is an interior point of ω . Let S represent the sample covariance matrix obtained from a random sample of size $N = n + 1$ from a multivariate normal population with mean vector $\underline{0}$ and covariance matrix Σ_0 . We will regard the vector $\underline{\theta}$ as consisting of mathematical variables and $\Sigma = \Sigma(\underline{\theta})$ as a matrix function of $\underline{\theta}$.

Consider the generalized least squares function (see, e.g., Browne (1974), Lee and Jennrich (1979))

$$Q(\underline{\theta}) = 2^{-1} \text{tr} \{ (S - \Sigma)V \}^2,$$

where V is a weight matrix which converges in probability to a positive definite matrix. For most practical applications, V is chosen so that it converges to Σ^{-1} in probability. Hence, in this paper, we restrict ourselves to this subclass of generalized least squares functions. One example is $V = S^{-1}$. We define the constrained generalized least squares estimator $\tilde{\underline{\theta}}$ of $\underline{\theta}_0$ as the vector which satisfies $\underline{h}(\tilde{\underline{\theta}}) = \underline{0}$ and minimizes $Q(\underline{\theta})$. It follows from the first order necessary condition that there exists a vector $\tilde{\underline{\lambda}}' = (\tilde{\lambda}_1, \dots, \tilde{\lambda}_r)$ of Lagrange multipliers such that

$$\begin{aligned} \dot{Q}(\tilde{\underline{\theta}}) + \tilde{L}'\tilde{\underline{\lambda}} &= \underline{0} \\ \underline{h}(\tilde{\underline{\theta}}) &= \underline{0}, \end{aligned} \quad (1)$$

where $\dot{Q} = (\partial Q / \partial \theta_i)$ is the gradient vector of $Q(\underline{\theta})$, and $L = (\partial h_i / \partial \theta_j)$ is an $r \times q$ matrix of partial derivatives, with $\tilde{L} = L(\tilde{\underline{\theta}})$.

The constrained maximum likelihood estimator $\hat{\underline{\theta}}$ of $\underline{\theta}_0$, is defined as the vector which satisfies $\underline{h}(\hat{\underline{\theta}}) = \underline{0}$ and minimizes the function

$$F(\underline{\theta}) = \log |\Sigma| + \text{tr}(\Sigma \Sigma^{-1}) - \log |S| - p.$$

Similarly, from the first order necessary condition, there corresponds a vector of Lagrange multipliers $\hat{\underline{\lambda}}' = (\hat{\lambda}_1, \dots, \hat{\lambda}_r)$, such that

$$\begin{aligned} \dot{\mathbf{F}}(\underline{\theta}) + \hat{\mathbf{L}}' \hat{\underline{\lambda}} &= \underline{0} \\ \mathbf{h}(\hat{\underline{\theta}}) &= \underline{0}, \end{aligned} \quad (2)$$

where $\dot{\mathbf{F}} = (\partial \mathbf{F} / \partial \theta_j)$ is the gradient of $\mathbf{F}(\underline{\theta})$, and $\hat{\mathbf{L}} = \mathbf{L}(\hat{\underline{\theta}})$.

Asymptotic properties for constrained maximum likelihood estimation have been studied by Aitchison and Silvey (1958). Here, we will consider some asymptotic properties for constrained generalized least squares estimation.

The following regularity conditions are imposed; these conditions are satisfied for almost every real-life application.

- (a) All elements of the $p^2 \times q$ matrix $\Delta = (\partial \sigma_{ij} / \partial \theta_k)$ are continuous on an open neighbourhood W of $\underline{\theta}_0$ in Ω .
- (b) $\underline{\theta}_0$ is identified; that is, the statements $\Sigma(\underline{\theta}_0) = \Sigma(\underline{\theta}^*)$, $\mathbf{h}(\underline{\theta}^*) = \underline{0}$, $\underline{\theta}^* \in \Omega$ imply that $\underline{\theta}^* = \underline{\theta}_0$.
- (c) Σ_0 is positive definite.
- (d) All elements in the matrix $\mathbf{L}$ are continuous on W .
- (e) The $r \times q$ matrix $\mathbf{L}_0 = \mathbf{L}(\underline{\theta}_0)$ is of rank r .
- (f) $\underline{\theta}_0$ is a regular point of the matrix-valued function of $\underline{\theta}$,

$$\mathbf{B} = \mathbf{B}(\underline{\theta}) = \begin{pmatrix} \Delta \\ \mathbf{L} \end{pmatrix},$$

that is $\mathbf{B}(\underline{\theta})$ has same rank for all $\underline{\theta} \in W$.

In the sequel, we denote $\underline{s}$ and $\underline{\sigma}$ as column vectors which respectively consist of all elements of $\mathbf{S}$ and Σ taken column by column. The following known results (see, e.g., Bargman, (1967); Browne, (1974); and Lee & Jennrich, (1979)) are required:

- (3) $\text{tr} \mathbf{V} \dot{\Sigma}_i \mathbf{V} \dot{\Sigma}_j = \Delta_i' (\mathbf{V} \otimes \mathbf{V}) \Delta_j$;
- (4) $\dot{\mathbf{Q}}(\underline{\theta}) = -\Delta' (\mathbf{V} \otimes \mathbf{V}) (\underline{s} - \underline{\sigma})$;
- (5) $\ddot{\mathbf{Q}}(\underline{\theta})(i, j) = \text{tr} \mathbf{V} \dot{\Sigma}_i \mathbf{V} \dot{\Sigma}_j - \text{tr} \mathbf{V} (\mathbf{S} - \Sigma) \mathbf{V} \ddot{\Sigma}_{ij}$;
- (6) $\dot{\mathbf{F}}(\underline{\theta}) = -\Delta' (\Sigma^{-1} \otimes \Sigma^{-1}) (\underline{s} - \underline{\sigma})$;

$$(7) \quad \ddot{F}(\underline{\theta})(i, j) = \text{tr} \Sigma^{-1} \dot{\Sigma}_i \Sigma^{-1} \dot{\Sigma}_j - \text{tr} \Sigma^{-1} (\mathbf{S} - \Sigma) \Sigma^{-1} (\ddot{\Sigma}_{ij} - 2\dot{\Sigma}_i \Sigma^{-1} \dot{\Sigma}_j);$$

$$\begin{aligned} \text{where} \quad \dot{\Sigma}_i &= (\partial \Sigma / \partial \theta_i), \\ \ddot{\Sigma}_{ij} &= (\partial^2 \Sigma / \partial \theta_i \partial \theta_j), \\ \ddot{Q}_{ij} &= (\partial^2 Q / \partial \theta_i \partial \theta_j), \\ \ddot{F}_{ij} &= (\partial^2 F / \partial \theta_i \partial \theta_j), \end{aligned}$$

Δ_i is the i th column of Δ , and $\otimes$ is the Kronecker product of matrices. Furthermore, we define

$$(8) \quad \mathbf{M}(\underline{\theta})(i, j) = \text{tr} \Sigma^{-1} \dot{\Sigma}_i \Sigma^{-1} \dot{\Sigma}_j,$$

which is equal to two times the information matrix of $\underline{\theta}$.

Proposition 1

The generalized least squares estimator $\tilde{\underline{\theta}}$ is consistent.

Proof:

Use of the implicit function theorem (see, e.g., Apostol, (1974)) together with assumptions (d) and (e) shows that there exists a $(q - r)$ subvector $\underline{\theta}^*$ of $\underline{\theta}$ such that all elements of $\underline{\theta}$ are continuous functions of $\underline{\theta}^*$ on a neighbourhood of $\underline{\theta}_0^*$. Since the composite of continuous functions is continuous, Σ is a continuous function of $\underline{\theta}^*$ on the neighbourhood of $\underline{\theta}_0^*$. The result now follows from Proposition 1 of Browne (1974). ||

Lemma 1. (Rothenberg, (1971))

The matrix $B_0 = B(\underline{\theta}_0)$ is of full column rank q .

Proof:

For any $\underline{\theta} \in W$, by Taylor's theorem and assumptions (a) and (d), we have

$$\underline{\sigma} = \underline{\sigma}_0 + \Delta^*(\underline{\theta} - \underline{\theta}_0),$$

and

$$\underline{h}(\underline{\theta}) = \underline{0} = L^*(\underline{\theta} - \underline{\theta}_0),$$

where $\underline{\sigma}_0 = \underline{\sigma}(\underline{\theta}_0)$, $\Delta^*(i, j) = \Delta(\underline{\theta}_i^*)(i, j)$, $L^*(r, s) = L(\underline{\theta}_r^*)(r, s)$, and all $\underline{\theta}_i^*$ and $\underline{\theta}_r^*$ lie on the line segment joining $\underline{\theta}$ and $\underline{\theta}_0$. Hence,

$$B^*(\underline{\theta} - \underline{\theta}_0) = \begin{pmatrix} \Delta^* \\ L^* \end{pmatrix} (\underline{\theta} - \underline{\theta}_0) = \begin{pmatrix} \underline{\sigma} - \underline{\sigma}_0 \\ 0 \end{pmatrix}.$$

It follows from assumption (b) that $B^*(\underline{\theta} - \underline{\theta}_0) = 0$ only if $\underline{\theta} = \underline{\theta}_0$. Thus B^* must be of full column rank. From assumption (f), B_0 must also be of full column rank. ||

Lemma 2

The matrix $\ddot{M}_0 = M_0 + L_0' L_0$ is nonsingular, where $M_0 = M(\underline{\theta}_0)$.

Proof:

From (3) and (8),

$$(9) \quad M_0 = \Delta_0' (\Sigma_0^{-1} \otimes \Sigma_0^{-1}) \Delta_0.$$

Let

$$\Phi_0 = \begin{pmatrix} \Sigma_0^{-1} \otimes \Sigma_0^{-1} & 0 \\ 0 & I_r \end{pmatrix},$$

with I_r denoting the identity matrix of order r . Then $\ddot{M}_0 = B_0' \Phi_0 B_0$. Since Φ_0 is nonsingular and B_0 is of full column rank by Lemma 2, it follows that $\ddot{M}_0$ is nonsingular. ||

Lemma 3

The matrix

$$\begin{pmatrix} M_0 & L_0' \\ L_0 & 0 \end{pmatrix}$$

is nonsingular and its inverse is

$$\begin{pmatrix} M_0 & L_0' \\ L_0 & 0 \end{pmatrix}^{-1} = \begin{pmatrix} P_0 & T_0 \\ T_0' & R_0 \end{pmatrix}$$

where

$$(10) \quad P_o = \ddot{M}_o^{-1} - \ddot{M}_o^{-1} L_o' (L_o \ddot{M}_o^{-1} L_o')^{-1} L_o \ddot{M}_o^{-1}$$

$$(11) \quad T_o = \ddot{M}_o^{-1} L_o' (L_o \ddot{M}_o^{-1} L_o')^{-1}$$

$$(12) \quad R_o = I_r - (L_o \ddot{M}_o^{-1} L_o')^{-1}$$

Proof:

The nonsingularity of $\ddot{M}_o$ follows from Lemma 2 and that of $L_o \ddot{M}_o^{-1} L_o'$ from assumption (e). The lemma then follows from Pringle & Rayner (1971), theorem 3.5, and the fact that a matrix with a unique conditional inverse is nonsingular. ||

It should be noted that examination of the nonsingularity of B_o and $\ddot{M}_o$ will give an idea as to whether or not $\underline{\theta}_o$ is identified. If the augmented matrix in Lemma 3 is singular, then $\underline{\theta}_o$ is not identified and/or some constraints are redundant so that L_o is of rank less than r . Following similar arguments as in McDonald & Krane (1977), it can be shown that the rank deficiency of B_o or of $\ddot{M}_o$ will be equal to the number of additional independent constraints required to achieve local identification. In practice, since B_o and $\ddot{M}_o$ are unknown, caution must be exercised when $B(\underline{\tilde{\theta}})$ and $\ddot{M}(\underline{\tilde{\theta}})$ are used (McDonald & Krane, (1979)).

The next lemma follows immediately from Pringle & Rayner (1971), corollary to theorem 3.5.

Lemma 4

The matrix

$$\begin{pmatrix} M_o + L_o L_o' & L_o' \\ L_o & 0 \end{pmatrix}$$

is nonsingular and its inverse is

$$\begin{pmatrix} M_o + L_o L_o' & L_o' \\ L_o & 0 \end{pmatrix}^{-1} = \begin{pmatrix} P_o & T_o \\ T_o' & R_o - I_r \end{pmatrix}.$$

Proposition 2

The joint asymptotic distribution of random variables $n^{1/2}(\underline{\tilde{\theta}} - \underline{\theta}_o)$ and $n^{1/2}\underline{\tilde{\lambda}}$ is multivariate normal with zero mean vector and covariance matrix

$$2 \begin{pmatrix} P_0 & 0 \\ 0 & -R_0 \end{pmatrix},$$

where P_0 and R_0 are given by (10) and (12) in Lemma 3. Alternatively, these matrices may be obtained via Lemma 4.

Proof:

Since $\tilde{\theta}$ is a consistent estimator of θ_0 , and θ_0 is an interior point of W , the probability that $\tilde{\theta}$ satisfies (1) tends to 1 as n tends to infinity.

From (1) and (4), we have

$$-\tilde{\Delta}'(V \otimes V)(\underline{s} - \underline{\sigma}) + \tilde{L}'\tilde{\lambda} = \underline{0},$$

where $\tilde{L} = L(\tilde{\theta})$, $\tilde{\Delta} = \Delta(\tilde{\theta})$, and $\tilde{\sigma} = \sigma(\tilde{\theta})$. Equivalently,

$$(13) \quad -\tilde{\Delta}'(V \otimes V) \{ \underline{s} - \underline{\sigma}_0 - (\tilde{\sigma} - \underline{\sigma}_0) \} + \tilde{L}'\tilde{\lambda} = \underline{0}.$$

By Taylor's theorem, there exist vectors θ_{-i}^* , $i = 1, \dots, q$ such that they lie on the line segment joining $\tilde{\theta}$ and θ_{-0} and

$$(14) \quad \tilde{\sigma} - \underline{\sigma}_0 = \Delta^*(\tilde{\theta} - \theta_{-0}),$$

where $\Delta^*(i, j) = \Delta(\theta_{-i}^*)(i, j)$. Hence, from (13),

$$(15) \quad \begin{aligned} &\tilde{\Delta}'(V \otimes V)\Delta^*(\tilde{\theta} - \theta_{-0}) + \tilde{L}'\tilde{\lambda} \\ &= \tilde{\Delta}'(V \otimes V)(\underline{s} - \underline{\sigma}_0). \end{aligned}$$

Similarly, by Taylor's theorem,

$$(16) \quad \underline{h}(\tilde{\theta}) = \underline{h}(\theta_{-0}) + L^*(\tilde{\theta} - \theta_{-0}),$$

where $L^*(r, s) = L(\theta_{-r}^*)(r, s)$ and all θ_{-r}^* lie on the line segment joining $\tilde{\theta}$ and θ_{-0} . From (16),

$$(17) \quad L^*(\tilde{\theta} - \theta_{-0}) = \underline{0}.$$

Since $\tilde{\theta}$ is a consistent estimator of θ_{-0} , $\frac{\theta_{-i}^*}{-i}$ and $\frac{\theta_{-1}^*}{-1}$ converge to θ_{-0} in probability as n tends to infinity. Consequently, it follows from the continuity of all first derivatives involved that $\tilde{\Delta}$ and $\tilde{\Delta}^*$ converge to Δ_0 in probability, and $\tilde{L}$ and L^* converge to L_0 in probability as n tends to infinity. Furthermore, V converges to Σ_0^{-1} in probability. Thus, it follows from (9), (15) and (17) that

$$(18) \quad \begin{pmatrix} M_0 & L_0' \\ L_0 & 0 \end{pmatrix} \begin{pmatrix} n^{1/2}(\tilde{\theta} - \theta_{-0}) \\ n^{1/2}\tilde{\lambda} \end{pmatrix} \underset{a}{=} \begin{pmatrix} n^{1/2}\Delta_0'(\Sigma_0^{-1} \otimes \Sigma_0^{-1})(\underline{s} - \underline{\sigma}_0) \\ 0 \end{pmatrix}.$$

By Lemma 3,

$$(19) \quad \begin{pmatrix} n^{1/2}(\tilde{\theta} - \theta_{-0}) \\ n^{1/2}\tilde{\lambda} \end{pmatrix} \underset{a}{=} \begin{pmatrix} P_0 & T_0 \\ T_0' & R_0 \end{pmatrix} \begin{pmatrix} n^{1/2}\Delta_0'(\Sigma_0^{-1} \otimes \Sigma_0^{-1})(\underline{s} - \underline{\sigma}_0) \\ 0 \end{pmatrix}.$$

Since from Browne (1974) the asymptotic distribution of $n^{1/2}\Delta_0'(\Sigma_0^{-1} \otimes \Sigma_0^{-1})(\underline{s} - \underline{\sigma}_0)$ is normal with mean vector $\underline{0}$ and covariance matrix $2M_0$, it follows that $n^{1/2}(\tilde{\theta} - \theta_{-0})$ and $n^{1/2}\tilde{\lambda}$ are jointly asymptotically normally distributed with a null mean vector and covariance matrix

$$2 \begin{pmatrix} P_0 & T_0 \\ T_0' & R_0 \end{pmatrix} \begin{pmatrix} M_0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} P_0 & T_0 \\ T_0' & R_0 \end{pmatrix}.$$

By arguments similar to those of Aitchison and Silvey (1958), it can be shown that the above matrix product is equal to

$$2 \begin{pmatrix} P_0 & 0 \\ 0 & -R_0 \end{pmatrix}.$$

The proof of the proposition is completed. ||

This proof is based on an adaptation by Swain (1975) to covariance structures of a proof of Rao (1973, p. 361) concerned with the estimation of parameters in the multinomial distribution.

It follows from Proposition 2 that $n^{1/2}(\tilde{\theta} - \theta_{-0})$ and $n^{1/2}\tilde{\lambda}$ are asymptotically inde-

pendent. The standard error of $\tilde{\theta}_i$ can be obtained from the square root of $2n^{-1}P_0(i, i)$.

Proposition 3

The generalized least squares estimator $(\tilde{\theta}, \tilde{\lambda})$ is asymptotically equivalent to the maximum likelihood estimator $(\hat{\theta}, \hat{\lambda})$.

Proof:

Following similar reasoning as in the proof of Proposition 2, it can be shown that

$$(20) \quad \begin{pmatrix} M_0 & L_0' \\ L_0 & 0 \end{pmatrix} \begin{pmatrix} n^{1/2}(\hat{\theta} - \theta_0) \\ n^{1/2}\hat{\lambda} \end{pmatrix} \underset{a}{=} \begin{pmatrix} n^{1/2}\Delta_0'(\hat{\Sigma}^{-1} \otimes \hat{\Sigma}^{-1})(\underline{s} - \underline{\sigma}_0) \\ 0 \end{pmatrix},$$

where $\hat{\Sigma} = \Sigma(\hat{\theta})$. From (15), (17) and (20),

$$(21) \quad \begin{pmatrix} n^{1/2}(\hat{\theta} - \tilde{\theta}) \\ n^{1/2}(\hat{\lambda} - \tilde{\lambda}) \end{pmatrix} \underset{a}{=} \begin{pmatrix} P_0 & T_0 \\ T_0' & R_0 \end{pmatrix} \begin{pmatrix} \Delta_0' \{ (V \otimes V) - (\hat{\Sigma}^{-1} \otimes \hat{\Sigma}^{-1}) \} n^{1/2}(\underline{s} - \underline{\sigma}_0) \\ 0 \end{pmatrix}.$$

Since $(V \otimes V) - (\hat{\Sigma}^{-1} \otimes \hat{\Sigma}^{-1})$ converges to zero in probability and $n^{1/2}(\underline{s} - \underline{\sigma}_0)$ converges in distribution, the right-hand side of (21) converges to zero in probability. Hence $(\tilde{\theta}, \tilde{\lambda})$ is asymptotically equivalent to $(\hat{\theta}, \hat{\lambda})$. $\parallel$

Lemma 5

The rank of P_0 is equal to $q-r$.

Proof:

Since $L_0 P_0 = 0$ and rank of L_0 is r , the rank of P_0 is less than or equal to $q-r$. On the other hand,

$$q = \text{rank}(M_0 P_0 + L_0' T_0') \leq \text{rank}(P_0) + r. \parallel$$

Before going on to the next proposition, we need one more notation from Browne (1974). Let $\underline{s}$ denote the vector obtained by taking the upper triangular elements including the diagonal of a symmetric matrix S , and let $\underline{g}$ denote the corresponding vector obtained from Σ . It is known that

$$\underline{s} = K_p' \underline{g},$$

where K_p is of order $p^2 \times p(p+1)/2$ with typical elements

$$K_p(ij, gh) = 2^{-1}(\delta_{ig}\delta_{jh} + \delta_{ih}\delta_{jg}), 1 \leq p, j \leq p, g \leq h \leq p,$$

and δ_{ig} represents the Kronecker's delta. This and other properties of K_p have been given by Browne (1974).

Proposition 4

The asymptotic distribution of $nQ(\tilde{\theta})$ is chi-square with degrees of freedom $p(p+1)/2 - (q-r)$.

Proof:

From (14), we have

$$\underline{s} - \underline{\tilde{\sigma}} = \underline{s} - \underline{\sigma}_0 - \Delta^*(\underline{\tilde{\theta}} - \underline{\theta}_0).$$

Hence, from (19) and the fact Δ^* converges to Δ_0 in probability as n tends to infinity, we have

$$(22) \quad n^{1/2}(\underline{s} - \underline{\tilde{\sigma}}) \stackrel{a}{=} \{I_p - \Delta_0 P_0 \Delta_0' (\Sigma_0^{-1} \otimes \Sigma_0^{-1})\} n^{1/2}(\underline{s} - \underline{\sigma}_0).$$

Now,

$$nQ(\tilde{\theta}) = 2^{-1} n(\underline{s} - \underline{\tilde{\sigma}})' (V \otimes V) (\underline{s} - \underline{\tilde{\sigma}}).$$

Using (22), it follows that

$$(23) \quad nQ(\tilde{\theta}) \stackrel{a}{=} 2^{-1} n(\underline{s} - \underline{\sigma}_0)' G_0 (\underline{s} - \underline{\sigma}_0),$$

where

$$G_0 = K_p^- \{I_p - (\Sigma_0^{-1} \otimes \Sigma_0^{-1}) \Delta_0 P_0 \Delta_0'\} (\Sigma_0^{-1} \otimes \Sigma_0^{-1}) \\ \cdot \{I_p - (\Sigma_0^{-1} \otimes \Sigma_0^{-1}) \Delta_0 P_0 \Delta_0'\}' K_p^{-1},$$

with K_p^- denoting the left inverse of K_p .

Since $n^{1/2}(\underline{s} - \underline{\sigma}_0)$ converges in distribution to $N(0, 2K_p'(\Sigma_0 \otimes \Sigma_0)K_p)$, see e.g. Browne (1974), and since $G_0 K_p'(\Sigma_0 \otimes \Sigma_0)K_p$ is an idempotent matrix, the asymptotic distribution of $nQ(\tilde{\theta})$ is chi-square with degrees of freedom equal to the rank of

$G_0 K_p' (\Sigma_0 \otimes \Sigma_0) K_p$ (Graybill, 1961, p. 83). Now,

$$\begin{aligned}
 \text{rank } \{G_0 K_p' (\Sigma_0 \otimes \Sigma_0) K_p\} &= \text{tr } \{G_0 K_p' (\Sigma_0 \otimes \Sigma_0) K_p\} \\
 &= \text{tr} K_p^- \{I_{p^2} - (\Sigma_0^{-1} \otimes \Sigma_0^{-1}) \Delta_0 P_0 \Delta_0'\} K_p \\
 &= \text{tr} I_{p(p+1)/2} - \text{tr} (\Sigma_0^{-1} \otimes \Sigma_0^{-1}) \Delta_0 P_0 \Delta_0' K_p K_p^- \\
 &= p(p+1)/2 - \text{tr} \Delta_0' (\Sigma_0^{-1} \otimes \Sigma_0^{-1}) \Delta_0 P_0 \\
 &= p(p+1)/2 - \text{tr} M_0 P_0,
 \end{aligned}$$

since $\Delta_0' K_p K_p^- = \Delta_0'$. Further, $M_0 P_0$ is an idempotent matrix,

$$\text{tr} M_0 P_0 = \text{rank } M_0 P_0 = \text{rank } \ddot{M}_0 P_0 = \text{rank } P_0 = q-r,$$

by Lemma 5. Hence the rank of $G_0 K_p' (\Sigma_0 \otimes \Sigma_0) K_p$ is $p(p+1)/2 - (q-r)$. ||

This proof is an adaptation of a proof of Browne (1974, p. 12).

Suppose $\underline{h}(\underline{\theta}) = (\underline{h}^*(\underline{\theta}), \underline{h}^{**}(\underline{\theta}))$, where $\underline{h}^*(\underline{\theta}) = (h_1(\underline{\theta}), \dots, h_j(\underline{\theta}))$, $\underline{h}^{**}(\underline{\theta}) = (h_{j+1}(\underline{\theta}), \dots, h_r(\underline{\theta}))$, and $j \leq r$. We assume that $\underline{\theta}_0$ remain identified when the last $(r-j)$ $h_i(\underline{\theta})$ are discarded. Let $\tilde{\theta}^*$ be the generalized least squares estimator that is subject to $\underline{h}^*(\underline{\theta}) = \underline{0}$, and $Q(\tilde{\theta}^*)$ be its final function value. The asymptotic distribution of $n \{Q(\tilde{\theta}) - Q(\tilde{\theta}^*)\}$ is given by the following proposition.

Proposition 5

The asymptotic distribution of $n \{Q(\tilde{\theta}) - Q(\tilde{\theta}^)\}$ is chi-square with degrees of freedom $r-j$.*

Proof:

Consider the partition $L_0' = (L_0^{*'}, L_0^{**'})$, analogous to $\underline{h}(\underline{\theta})$, and let $\ddot{M}_0^* = M_0 + L_0^{*'} L_0^*$. From arguments similar to those in the proof of Proposition 4, it can be shown that

$$(24) \quad nQ(\tilde{\theta}^*) \stackrel{a}{=} 2^{-1} n^{1/2} (\underline{s} - \underline{g}_0)' G_0^* (\underline{s} - \underline{g}_0),$$

where

$$\begin{aligned}
G_o^* &= K_p^- \{I_{p^2} - (\Sigma_o^{-1} \otimes \Sigma_o^{-1}) \Delta_o P_o^* \Delta_o'\} (\Sigma_o^{-1} \otimes \Sigma_o^{-1}) \\
&\quad \cdot \{I_{p^2} - (\Sigma_o^{-1} \otimes \Sigma_o^{-1}) \Delta_o P_o^* \Delta_o'\}' K_p^{-'}, \\
P_o^* &= \bar{M}_o^{*-1} - \bar{M}_o^{*-1} L_o^{*'} (L_o^* \bar{M}_o^{*-1} L_o^{*'})^{-1} L_o^* \bar{M}_o^{*-1}.
\end{aligned}$$

The matrix P_o^* possesses similar properties as P_o and its rank is equal to $q-j$. From (23) and (24)

$$n \{Q(\tilde{\theta}) - Q(\tilde{\theta}^*)\} \stackrel{a}{\approx} 2^{-1} n^{1/2} (\underline{s} - \underline{g}_o)' (G_o - G_o^*) (\underline{s} - \underline{g}_o).$$

The matrix $(G_o - G_o^*) K_p' (\Sigma_o \otimes \Sigma_o) K_p$ is idempotent, with its rank equal to

$$\begin{aligned}
&\text{tr}(G_o - G_o^*) K_p' (\Sigma_o \otimes \Sigma_o) K_p \\
&= \text{tr} K_p^- (\Sigma_o^{-1} \otimes \Sigma_o^{-1}) \Delta_o (P_o^* - P_o) \Delta_o' (\Sigma_o^{-1} \otimes \Sigma_o^{-1}) K_p^- K_p' (\Sigma_o \otimes \Sigma_o) K_p \\
&= \text{tr} (\Sigma_o^{-1} \otimes \Sigma_o^{-1}) \Delta_o (P_o^* - P_o) \Delta_o' K_p^- K_p^- \\
&= \text{tr} M_o (P_o^* - P_o) \\
&= \text{rank } P_o^* - \text{rank } P_o \\
&= r-j.
\end{aligned}$$

The proof of the proposition is completed. ||

Proposition 4 provides an asymptotic test statistic for testing the null hypothesis

$$(25) \quad H_o = \Sigma_o = \Sigma(\underline{\theta}_{-o}), \underline{\theta}_{-o} \in \omega$$

against the general alternative that Σ_o is any symmetric positive definite $p \times p$ matrix. The validity of various constraints can be assessed by means of Proposition 5. Another asymptotic test statistic for the null hypothesis (25) against the specific alternative

$$H_1 = \Sigma = \Sigma(\underline{\theta}_{-o}), \underline{\theta}_{-o} \in \Omega$$

is given by the next proposition.

Proposition 6

The asymptotic distribution of $-2^{-1} \underline{n} \tilde{\lambda}' \tilde{R}^{-1} \tilde{\lambda}$ under H_0 is chi-square with degrees of freedom equal to the rank of R_0 , where

$$\tilde{R} = I_r - (\tilde{L} M(\tilde{\theta})^{-1} \tilde{L}')^{-1} \text{ (see Lemma 3).}$$

Proof:

From the regularity assumptions,

$$-2^{-1} \underline{n} \tilde{\lambda}' \tilde{R}^{-1} \tilde{\lambda} = -2^{-1} \underline{n} \tilde{\lambda}' R_0^{-1} \tilde{\lambda}.$$

By Proposition 2, the asymptotic distribution $\underline{n}^{1/2} \tilde{\lambda}$ is $N(0, -2R_0)$. The proposition follows from the fact that $R_0^{-1} R_0$ is an idempotent matrix of rank equal to that of R_0 . ||

The analogous result on $\tilde{\lambda}$ has been reported by Silvey (1959).

For most covariance structure models, the constrained generalized least squares estimator and the Lagrangian multiplier $\tilde{\lambda}$ cannot be solved in closed form; thus some non-linear iterative procedure has to be used. Lee (1980) has applied the penalty function technique (Fiocco & McCormick, (1968)) to obtain $(\tilde{\theta}, \tilde{\lambda})$ in a confirmatory factor analysis model. However, Browne (private communication) pointed out that if one can impose equality constraints by finding a reparameterization, it will result in a simpler and more efficient algorithm for obtaining estimates. Below, we point out relationships between the Lagrangian multiplier approach and the reparameterization approach. We need three additional lemmas.

Lemma 6

If $h(\underline{\theta}) = 0$ for all $\underline{\theta} \in \omega$, then there exists a $q \times 1$ vector-valued function $\underline{q}(\underline{\gamma})$ and a $(q-r) \times 1$ parameter vector $\underline{\gamma}$ such that $\underline{\theta} = \underline{g}(\underline{\gamma}) \in \Omega$. Further, $\underline{g}(\underline{\gamma})$ is continuous on a neighbourhood of $\underline{\gamma}_0$, where $\underline{g}(\underline{\gamma}_0) = \underline{\theta}_0$.

Proof:

From Proposition 1, there exists $\underline{\theta}^*$ and a $q \times 1$ continuous function f_1 , such that $\underline{\theta} = f_1(\underline{\theta}^*)$. Let $\underline{\gamma} = f_2(\underline{\theta}^*)$ be any one to one continuous function of $\underline{\theta}^*$. Then $\underline{\theta} = f_1\{f_2^{-1}(\underline{\gamma})\} = \underline{g}(\underline{\gamma})$. The function of $\underline{g}$ is continuous since both f_1 and f_2^{-1} are continuous. ||

Lemma 7

Let $K = \partial \underline{g}(\underline{\gamma}) / \partial \underline{\gamma}' = (\partial g_i(\underline{\gamma}) / \partial \gamma_j)$ be the $q \times (q-r)$ matrix-valued function of $\underline{\gamma}_0$.

Then for any $\underline{\theta} \in \omega$, $LK = \underline{0}$.

Proof:

From the chain rule of vectors,

$$\underline{0} = \frac{\partial \underline{0}}{\partial \underline{\gamma}'} = \frac{\partial \underline{h}}{\partial \underline{\gamma}'} = \frac{\partial \underline{h}}{\partial \underline{\theta}'} \cdot \frac{\partial \underline{g}}{\partial \underline{\gamma}'} = LK. \parallel$$

Lemma 8 (Khatri, 1966):

Suppose that

- (a) L is an $r \times q$ matrix of rank r ,
- (b) K is a $q \times (q-r)$ matrix of rank $(q-r)$,
- (c) $LK = \underline{0}$,
- (d) C is a $q \times q$ positive definite matrix.

Then

$$K(K'CK)^{-1}K' = C^{-1} - C^{-1}L'(LC^{-1}L')^{-1}LC^{-1}.$$

Now let $\tilde{\underline{\gamma}}$ be the unconstrained generalized least squares estimator of $\underline{\gamma}_0$ obtained by minimizing

$$Q(\underline{\gamma}') = 2^{-1} \text{tr} \{ [S - \Sigma(\underline{g}(\underline{\gamma}))]' V]^2,$$

and let

$$(26) \quad \tilde{\underline{\theta}} = \underline{g}(\tilde{\underline{\gamma}}).$$

From the definition of $\underline{g}(\tilde{\underline{\gamma}})$ in Lemma 5 we have $\underline{h}(\tilde{\underline{\theta}}) = \underline{0}$, and $\tilde{\underline{\theta}}$ is the generalized least squares estimator of $\underline{\theta}_0$ subject to the constraints $\underline{h}(\underline{\theta}) = \underline{0}$.

Since, by the chain rule for vectors,

$$\left. \frac{\partial \underline{\sigma}}{\partial \underline{\gamma}'} \right|_{\underline{\gamma} = \underline{\gamma}_0} = \left. \frac{\partial \underline{\sigma}}{\partial \underline{\theta}} \frac{\partial \underline{\theta}}{\partial \underline{\gamma}'} \right|_{\underline{\gamma} = \underline{\gamma}_0} = \Delta_0 K_0,$$

it follows from Proposition 2 of Browne (1974) that the limiting distribution of $n^{1/2}(\tilde{\underline{\gamma}} - \underline{\gamma}_0)$ is normal with a null mean vector and covariance matrix

$$2 \{ K_0' \Delta_0' (\Sigma_0^{-1} \otimes \Sigma_0^{-1}) \Delta_0 K_0 \}^{-1}$$

$$= 2(K_O' \ddot{M}_O K_O)^{-1}$$

$$= 2(K_O' \ddot{M}_O K_O)^{-1},$$

from Lemma 6.

In the usual manner, using a first order Taylor series approximation, it can be shown that

$$n^{1/2}(\tilde{\theta} - \theta_O) \stackrel{a}{=} n^{1/2} K_O (\tilde{\gamma} - \gamma_O),$$

so that the asymptotic covariance matrix of $n^{1/2}(\tilde{\theta} - \theta_O)$ is

$$2K_O(K_O' \ddot{M}_O K_O)^{-1} K_O'.$$

By Lemma 8, this covariance matrix is equal to

$$2 \{ \ddot{M}_O^{-1} - \ddot{M}_O^{-1} L_O (L_O' \ddot{M}_O^{-1} L_O)^{-1} L_O' \ddot{M}_O^{-1} \}.$$

This is the same result as that derived in Proposition 2.

Hence, if it is possible to find a γ such that $g(\gamma)$ is a simple closed form expression, the constrained generalized least squares estimates $\tilde{\theta}$ can be obtained by means of (26).

ACKNOWLEDGEMENT

The authors are deeply indebted to Professor M.W. Browne whose contributions nearly amounted to co-authorship.

REFERENCES

- AITCHISON, J. and SILVEY, D.S. (1958). Maximum likelihood estimation of parameters subject to restraints. *Ann. Math. Statist.*, **29**, 813-828.
- ANDERSON, T.W. (1973). Asymptotically efficient estimation of covariance matrices with linear structure. *Ann. Math. Statist.*, **1**, 135-141.
- APOSTOL, T.M. (1973). *Mathematical analysis, Second edition*. Reading, Mass.: Addison-Wesley.
- BARGMANN, R.E. (1967). Matrices and determinants. In S.M. Selby (Ed.), *Chemical Rubber Company Handbook of Tables for Mathematics*. Cleveland, Ohio: Chemical Rubber Company.
- BENTLER, P.M. (1980). Multivariate analysis with latent variables: Causal modeling. *Annual Rev. Psychol.*, **31**, 419-456.
- BENTLER, P.M. and WEEKS, D.G. (1980). Linear structural equations with latent variables. *Psychometrika*, **45**, 289-308.

- BROWNE, M.W. (1974). Generalized least squares estimators in the analysis of covariance structures. *South African Statistical Journal*, 8, 1-24.
- FIACCO, A.V. and McCORMICK, G.P. (1968). *Nonlinear programming: Sequential unconstrained minimization technique*. New York: Wiley.
- JÖRESKOG, K.G. (1973). Analysis of covariance structures. In P.R. Krishnaiah, (Ed.), *Multivariate Analysis-III*, New York: Academic.
- JÖRESKOG, K.G. (1978). Structural analysis of covariance and correlation matrices. *Psychometrika*, 43, 443-477.
- JÖRESKOG, K.G. and SÖRBOM, D. (1978). *LISREL IV User Guide*. Chicago: National Education Research.
- KHATRI, C.G. (1966). A note on a MANOVA model applied to problems in growth curves. *Ann. Inst. Statist. Math.*, 18, 75-86.
- LEE, S.Y. (1980). Estimation of covariance structure models with parameters subject to functional restraints. *Psychometrika*, 45, 309-324.
- LEE, S.Y. and JENNRICH, R.I. (1979). A study of algorithms for covariance structure analysis with specific comparisons using factor analysis. *Psychometrika*, 44, 99-113.
- MCDONALD, R.P. (1978). A simple comprehensive model for the analysis of covariance structures. *British Journal of Mathematical and Statistical Psychology*, 31, 59-72.
- MCDONALD, R.P. & KRANE, W.R. (1977). A note on local identifiability and degrees of freedom in the asymptotic likelihood ratio test. *British Journal of Mathematical and Statistical Psychology*, 30, 198-203.
- MCDONALD, R.P. & KRANE, W.R. (1979). A Monte Carlo study of local identifiability and degrees of freedom in the asymptotic likelihood ratio test. *British Journal of Mathematical and Statistical Psychology*, 32, 121-132.
- PRINGLE, R.M. & RAYNER, A. (1971). Generalized inverse matrices with applications to statistics. *Griffin Statistical Monograph Series # 28*. London: Griffin.
- RAO, C.R. (1973). *Linear Statistical Inference and its Applications*, Second Edition. New York: Wiley.
- ROTHENBERG, T.J. (1971). Identification in parametric models. *Econometrica*, 37, 577-591.
- SILVEY, D.S. (1959). The Lagrangian multiplier test. *Ann. Math. Statist.*, 30, 389-407.
- SWAIN, A.J. (1975). *Analysis of parametric structures for variance matrices*. Unpublished Ph.D. thesis, University of Adelaide.